



BAULKHAM HILLS HIGH SCHOOL

TRIAL 2014
YEAR 12 TASK 4

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-14
- Marks may be deducted for careless or badly arranged work

Total marks – 70

Exam consists of 11 pages.

This paper consists of TWO sections.

Section 1 – Page 2-4 (10 marks)
Questions 1-10

- Attempt Question 1-10

Section II – Pages 5-10 (60 marks)

- Attempt questions 11-14

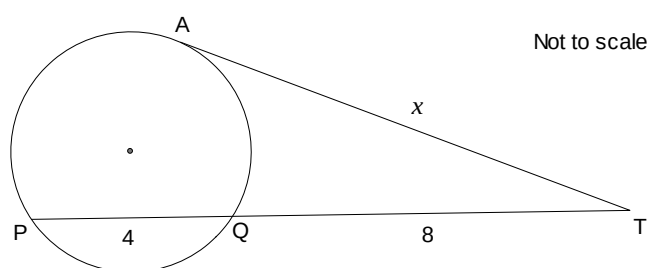
Table of Standard Integrals is on page 11

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. Given the equation $A = 10e^{-kt}$, what is the value of k given that $A = 3.6$ and $t = 5$.

(A) -0.717
(B) -0.204
(C) 0.204
(D) 0.717

2.



In the diagram above, TA is a tangent and PQ is a chord produced to T . The value of x is

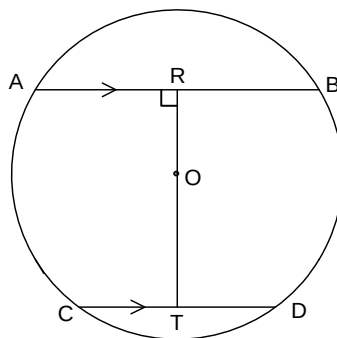
(A) 12
(B) $2\sqrt{3}$
(C) $4\sqrt{2}$
(D) $4\sqrt{6}$

3. How many distinct permutations of the letter of the word " $DIVIDE$ " are possible in a straight line when the word begins and ends with the letter D

(A) 12
(B) 180
(C) 360
(D) 720

4.	The coordinates of the point that divides the interval joining $(-7,5)$ and $(-1,-7)$ externally in the ratio 1:3 are (A) $(-10,8)$ (B) $(-10,11)$ (C) $(2,8)$ (D) $(2,11)$
5.	What is the domain and range of $y = 2 \cos^{-1} \frac{3x}{2}$? (A) $D = \left\{x: -\frac{2}{3} \leq x \leq \frac{2}{3}\right\}, R = \{y: 0 \leq y \leq 2\pi\}$ (B) $D = \left\{x: -\frac{3}{2} \leq x \leq \frac{3}{2}\right\}, R = \{y: 0 \leq y \leq 2\pi\}$ (C) $D = \left\{x: -\frac{2}{3} \leq x \leq \frac{2}{3}\right\}, R = \left\{y: 0 \leq y \leq \frac{\pi}{2}\right\}$ (D) $D = \left\{x: -\frac{3}{2} \leq x \leq \frac{3}{2}\right\}, R = \left\{y: 0 \leq y \leq \frac{\pi}{2}\right\}$
6.	Which of the following is the general solution of $3 \tan^2 x - 1 = 0$, where n is an integer? (A) $n\pi \pm \frac{\pi}{6}$ (B) $n\pi \pm \frac{\pi}{3}$ (C) $2n\pi \pm \frac{\pi}{6}$ (D) $2n\pi \pm \frac{\pi}{3}$
7.	The displacement of a particle moving in simple harmonic motion is given by $x = 3 \cos \pi t$ where t is the time in seconds. The period of oscillation is: (A) π (B) $\frac{2\pi}{3}$ (C) 2 (D) 3

8. AB and CD are parallel chords in a circle, which are 10cm apart.
 $OR \perp AB$, $AB = 14\text{cm}$ and $CD = 12\text{cm}$.



Find the diameter of the circle to 1 decimal place

- (A) 4.4cm
 (B) 8.2cm
 (C) 14.8cm
 (D) 16.5cm
9. The domain of $f(x) = \log_e[(x - 4)(5 - x)]$ is
- (A) $4 \leq x \leq 5$
 (B) $x \leq 4, x \geq 5$
 (C) $4 < x < 5$
 (D) $x < 4, x > 5$

10. Which of the following represents the derivate of $y = \sin^{-1}\left(\frac{1}{x}\right)$?

- (A) $\frac{1}{x\sqrt{x^2-1}}$
 (B) $\frac{1}{\sqrt{x^2-1}}$
 (C) $\frac{-1}{x\sqrt{x^2-1}}$
 (D) $\frac{-1}{\sqrt{x^2-1}}$

End of Section 1

Section II – Extended Response**All necessary working should be shown in every question.**

Question 11 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	Evaluate $\int_0^{\frac{\pi}{4}} \cos^2 4x \, dx$	3
b)	Find $\int \frac{dx}{x(\log_e x)^{11}}$, using the substitution $u = \log_e x$	2
c)	Prove the identity $\frac{1 + \sin 2x + \cos 2x}{\cos x + \sin x} = 2 \cos x$	2
d)	Solve for x $\frac{4}{x-1} \leq 3$	3
e)	(i) Show that a root of the continuous function $f(x) = x^3 - \ln(x+1)$ lies between 0.8 and 0.9.	1
	(ii) Hence use the halving the interval method to find the value of the root correct to 1 decimal place.	1
f)	(i) Find $\frac{d}{dx} \left[\tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) \right]$	2
	(ii) Hence sketch $y = \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right)$ for $-2 \leq x \leq 2$	1
End of Question 11		

Question 12 (15 marks) - Start on the appropriate page in your answer booklet

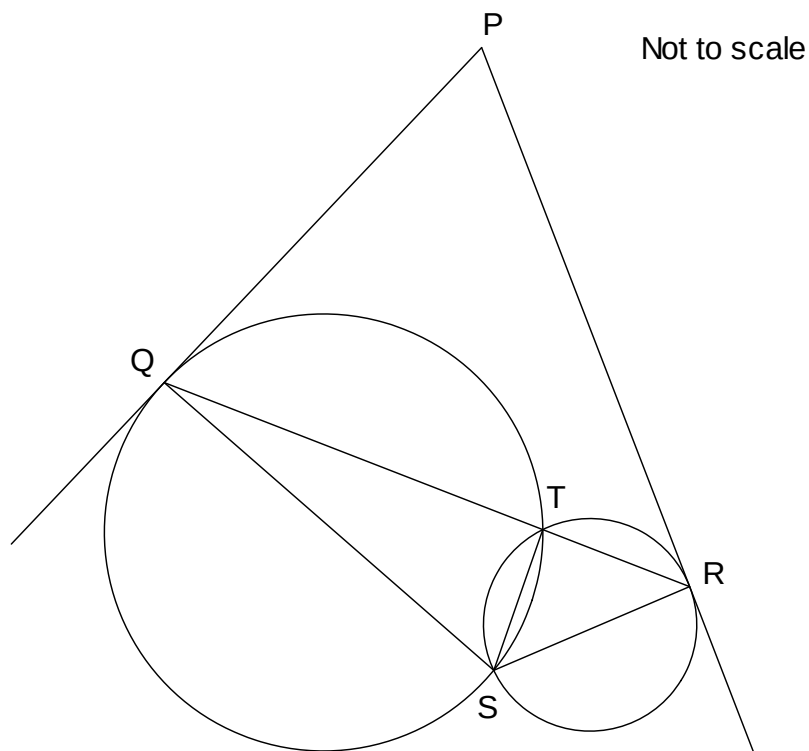
Marks

- a) When a polynomial $P(x)$ is divided by $x^2 - 4$ the remainder is $2x + 3$.
What is the remainder when $P(x)$ is divided by $x - 2$

2

- b) In the given diagram, PQ and PR are tangents and Q, T, R are collinear.

3

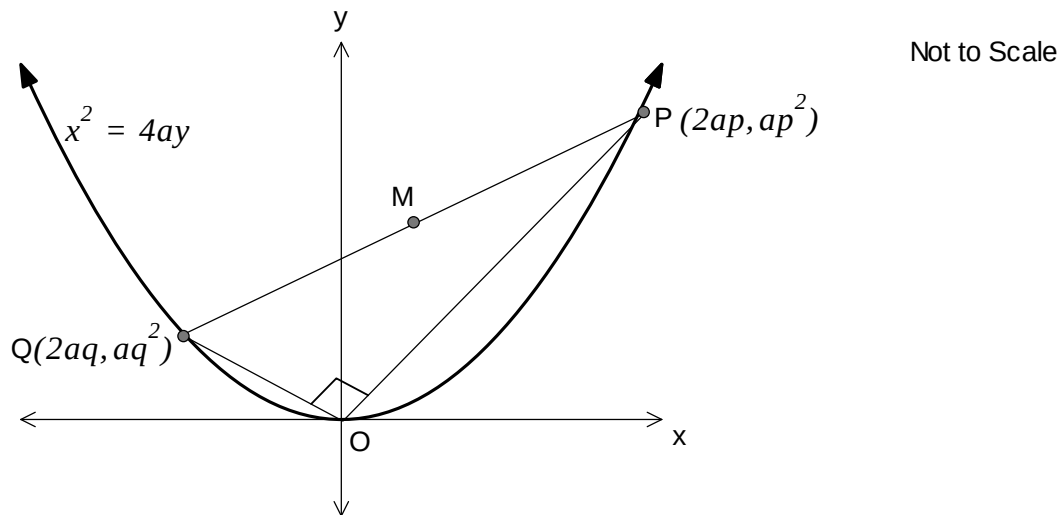


Copy or trace the diagram in to your writing booklet.
Prove that the points P, Q, S, R are concyclic.

Question 12 continues on the following page

Question 12 (continued)

c)



Points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.
The chord PQ subtends a right angle at the origin.

(i) Prove $pq = -4$

2

(ii) Find the equation of the locus of M , the midpoint of PQ .

3

d)

Find the coefficient of x^4 in the expression of $\left(x - \frac{2}{x}\right)^{12}$

2

e)

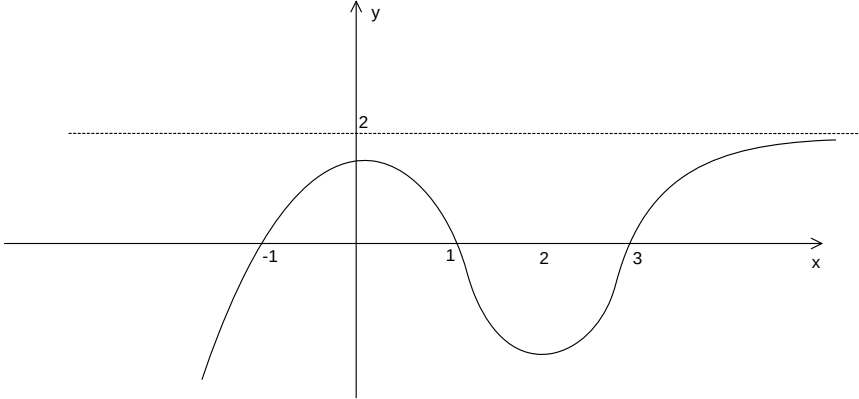
Prove by mathematical induction

$$1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \cdots + n \times 2^n = (n-1)2^{n+1} + 2$$

for positive integers $n \geq 1$

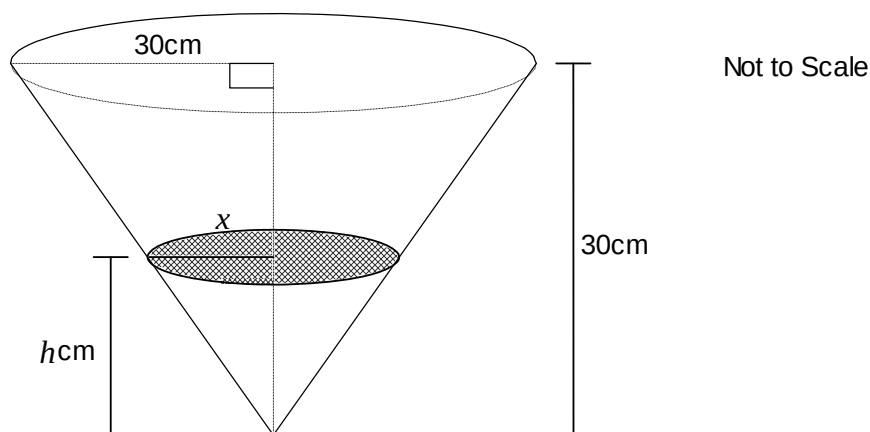
3

End of Question 12

Question 13 (15 marks) - Start on the appropriate page in your answer booklet		Marks
a)	(i) Express $\sqrt{3} \sin x - \cos x$ in the form $R \sin(x - \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. (ii) Hence state the least value of $\sqrt{3} \sin x - \cos x$ and the smallest positive value of x for this least value to occur.	2 2
b)	In the cubic equation $3x^3 - (2k - 4)x^2 + 5x + k^2 = 0$ the sum of the roots is equal to twice their product. Find the values of k .	3
c)	Find the number of arrangements of the letters of the word <i>PENCILS</i> if there are 3 letters between <i>E</i> and <i>I</i> .	2
d)	Below is the graph of a function $y = f(x)$  Copy the diagram in your booklet, and on the same set of axes sketch a possible graph for $y = f'(x)$.	2
e)	It is estimated that the rate of increase in the population of a particular species of bird is given by the equation $\frac{dP}{dt} = kP(L - P)$ where k and L are positive constants. (i) Verify that for any positive constant c, the expression $P = \frac{Lc}{c + e^{-kLt}}$ satisfies the above differential equation. (ii) What can be deduced about P as t increases?	3 1
End of Question 13		

Question 14 (15 marks) - Start on the appropriate page in your answer booklet

a)



Water is poured into a conical vessel at a constant rate of $24\text{cm}^3/\text{s}$.
The depth of water is $h\text{cm}$ at any time t seconds.

- | | | |
|-------|---|----------|
| (i) | Show that the volume of water is given by $V = \frac{1}{3}\pi h^3$. | 1 |
| (ii) | Find the rate at which the depth of water is increasing when $h = 16\text{cm}$. | 2 |
| (iii) | Hence find that rate of increase of the area of surface of the liquid when $h = 16$. | 1 |

- b) The acceleration of a particle is given by the equation $\frac{d^2x}{dt^2} = 8x(x^2 + 1)$, where x is the displacement in centimetres from a fixed point O , after t seconds.
Initially the particle is moving from O with speed 2cm/s in a negative direction.

- | | | |
|-------|---|----------|
| (i) | Prove the general result $\ddot{x} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$. | 2 |
| (ii) | Hence show that the speed is given by $2(x^2 + 1)\text{cm/s}$. | 2 |
| (iii) | Find an expression for x in terms of t . | 2 |

Question 14 continues on the following page

Question 14 (continued)

- c) A projectile is fired from the origin with velocity V with an angle of elevation θ , where $\theta \neq \frac{\pi}{2}$.

YOU MAY ASSUME

$$x = Vt \cos \theta, \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta$$

Where x and y are the horizontal and vertical displacements from O , t seconds after firing

- (i) Show the equation of flight can be expressed as

$$y = x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta) \quad \text{where } h = \frac{V^2}{2g} \quad \mathbf{2}$$

- (ii) Show that a point (X, Y) can be hit by firing at 2 different angles θ_1 and θ_2 provided $X^2 < 4h(h - Y)$. **2**

- (iv) Show that no point above the x -axis can be hit by firing at 2 different angles θ_1 and θ_2 satisfying both $\theta_1 < \frac{\pi}{4}$ and $\theta_2 < \frac{\pi}{4}$. **1**

End of Paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

EXT 1 SOLUTIONS

$$1) 3 \cdot 6 = 10 e^{-5K}$$

$$0.36 = e^{5K}$$

$$K = 0.204 \quad \boxed{C}$$

$$2) x^2 = 12 \times 8$$

$$= 96$$

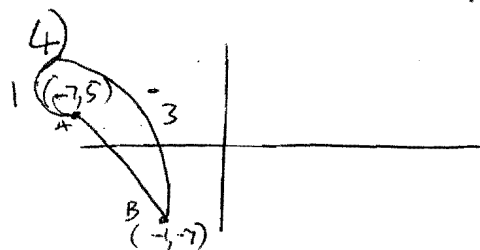
$$x = \sqrt{96}$$

$$= 4\sqrt{6} \quad \boxed{D}$$

$$3) \boxed{D} \quad \frac{4!}{2!} \quad \boxed{D}$$

$$\frac{4 \times 3 \times 2}{2} = 12$$

$$\boxed{A}$$



$$X = \frac{1 \times -1 + 3 \times -7}{-2}$$

$$= \frac{-1 + 21}{-2}$$

$$= -10$$

$$Y = \frac{1 \times 7 + 3 \times 5}{-2}$$

$$= \frac{7 + 15}{-2} = 11$$

$$\boxed{B}$$

$$5) y = 2 \cos^{-1} \frac{3x}{2}$$

$$\frac{y}{2} = \cos^{-1} \frac{3x}{2}$$

$$-1 \leq \frac{3x}{2} \leq 1$$

$$-\frac{2}{3} \leq x \leq \frac{2}{3}$$

$$0 \leq \frac{y}{2} \leq \pi$$

$$0 \leq y \leq 2\pi \quad \boxed{A}$$

$$6) 3 \tan^2 x - 1 = 0$$

$$\tan^2 x = \frac{1}{3}$$

$$\tan x = \pm \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6}, \frac{\pi}{2} - \frac{\pi}{6}, \pi + \frac{\pi}{6}$$

$$2\pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}$$

$$= n\pi \pm \frac{\pi}{6} \quad \boxed{A}$$

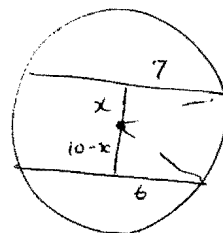
$$7) x = 3 \cos \pi t$$

$$\text{Period} = \frac{2\pi}{\pi}$$

$$= 2$$

$$\boxed{C}$$

8/



$$7^2 + x^2 = (10-x)^2 + 6^2$$

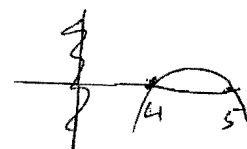
$$x = \frac{80}{27}$$

$$r = \sqrt{7^2 + \left(\frac{80}{27}\right)^2}$$

$$= 8.24 \dots$$

$$\therefore d = 16.5 \quad \boxed{D}$$

$$(9) (x-4)(5-x) > 0$$



$$4 < x < 5 \quad \boxed{C}$$

$$10. y = \sin^{-1}\left(\frac{1}{x}\right)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \cdot \frac{-1}{x^2}$$

$$= \frac{-1}{x^2 \sqrt{1 - \frac{1}{x^2}}}$$

$$= \frac{-1}{x \sqrt{x^2 - 1}} \quad \boxed{C}$$

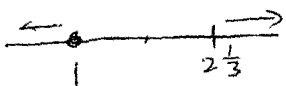
Q11

$$\begin{aligned} \cos 2x &= 2\cos^2 x - 1 \\ \cos^2 x &= \frac{1}{2} [\cos 2x + 1] \\ \cos^2 4x &= \frac{1}{2} [\cos 8x + 1] \end{aligned}$$

$$\begin{aligned} \int_0^{\pi/4} \cos^2 4x dx &= \frac{1}{2} \int_0^{\pi/4} \cos 8x + 1 dx \quad (1) \\ &= \frac{1}{2} \left[\frac{1}{8} \sin 8x + x \right]_0^{\pi/4} \quad (1) \\ &= \frac{1}{2} \left[\left(0 + \frac{\pi}{4}\right) - 0 \right] \\ &= \pi/8 \quad (1) \end{aligned}$$

$$\begin{aligned} c) \text{ LHS} &= \frac{1 + 2\sin x \cos x + 2\cos^2 x - 1}{\cos x + \sin x} \quad (1) \\ &= \frac{2\cos x (\sin x + \cos x)}{\cos x + \sin x} \\ &= 2\cos x \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} d) \frac{4}{x-1} &\leq 3 \\ 4 &\leq 3x-3 \\ \frac{7}{3} &\leq x \end{aligned}$$



$$\begin{aligned} b) \int \frac{dx}{x(\log x)^{10}} \\ u &= \log x \\ \frac{du}{dx} &= \frac{1}{x} dx \\ &= \int \frac{du}{u^{10}} \quad (1) \\ &= \int u^{-10} \\ &= \frac{u^{-10}}{-10} + C \\ &= -\frac{1}{10} (\log x)^{-10} + C \quad (1) \end{aligned}$$

$$\begin{aligned} x &< 1, x \geq 2\frac{1}{3} \\ \text{if } x &\leq 1, x \geq 2\frac{1}{3} \quad (2 \text{ marks}) \\ \text{if } 1 &< x \leq 2\frac{1}{3} \quad (2 \text{ marks}) \end{aligned}$$

$$11) i) f(x) = x^2 - \ln(x+1)$$

$$f(0.8) = -0.075$$

$$f(0.9) = 0.46$$

$\therefore f(0.8) \cdot f(0.9) < 0$ opp. sign $\therefore$ Root exists (1)

Let 1st APPROX $x = 0.85$.

$$f(0.85) = -0.001$$

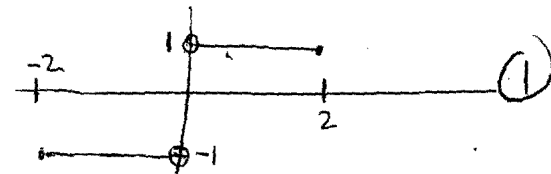
$\therefore$ Root lies between 0.85 & 0.9

$\therefore$ Root = 0.9 to 1 dec Place. (1)

11f)

$$\begin{aligned} c) \frac{d}{dx} \left[\tan^{-1} x + \tan^{-1} \frac{1}{x} \right] \\ &= \frac{1}{1+x^2} + \frac{1}{1+(\frac{1}{x})^2} \cdot \frac{-1}{x^2} \\ (1) \quad &= \frac{1}{1+x^2} - \frac{1}{x^2+1} \\ &= 0 \quad (1) \end{aligned}$$

$\therefore y = \tan^{-1} x + \tan^{-1}(\frac{1}{x})$ is horizontal



$$12a) P(x) = (x^2 - 4)(Q(x)) + 2x + 3$$

$$P(2) = 4 + 3 = 7$$

$$\therefore \text{Rem} = 7$$

$$b) \angle PRT = \angle TSR = x^\circ \quad \left(\begin{array}{l} \text{L between} \\ \text{trans \& chad} \end{array} \right)$$

$$\textcircled{1} = \text{L \& ALT segment}$$

SIMILARLY

$$\angle QAR = \angle QST = y^\circ$$

$$\angle QSR = x^\circ + y^\circ$$

$$\angle QPR = 180 - x - y \quad (\text{sum of } \Delta)$$

$$\therefore \angle QSR + \angle QPR = 180^\circ$$

$\therefore P, Q, S, R$ are concyclic.

$$c) (i) m_{OP} = \frac{q p^2 - 0}{2ap - 0}$$

$$= \frac{p}{2}$$

$$\text{Similarly } m_{OQ} = \frac{q}{2}$$

$$m_{OP} \times m_{OQ} = -1 \quad \text{since lines } \perp$$

$$\text{i.e. } \frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4$$

$$(ii) \text{Midpoint } \left(\frac{2ap + 2aq}{2}, \frac{q^2 + aq^2}{2} \right)$$

$$= \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

$$x = a(p+q)$$

$$\therefore p+q = \frac{x}{a} \quad y = a \left[\frac{(p+q)^2 - 2pq}{2} \right]$$

$$y = a \left[\frac{\frac{x^2}{a^2} - 2(-4)}{2} \right]$$

$$y = \frac{x^2}{2a} + 4a$$

or equivalent for $n=k$. $\therefore$ true for $n=1, n=2 \rightarrow$ for all n by $\textcircled{1}$

$$12d) T_{K+1} = C_K x^K \left(\frac{-2}{x} \right)^{12-K} \quad \textcircled{1}$$

$$= {}^{12}C_K x^K (-2)^{12-K} x^{K-12}$$

$$= {}^{12}C_K (-2)^{12-K} x^{2K-12}$$

$$\therefore 2K - 12 = 4$$

$$K = 8$$

$$\therefore \text{coeff} = {}^{12}C_8 (-2)^4 = 7920 \quad \textcircled{1}$$

e) Prove True $N=1$

$$\text{I.e. } 1 \times 2 = (1-1)2^2 + 2$$

$$= 2 \quad \text{yes.} \quad \textcircled{1}$$

2 ticks - 1 mark

3 ticks - 2 marks

4 ticks - 3 marks

ASSUME TRUE $N=K$.

$$1 \times 2 + 2 \times 2^2 + 3 \times 2^3 + \dots + K \times 2^K = (K-1)2^{K+1} + 2$$

Prove True $N=K+1$.

$$1 \times 2 + \dots + K \times 2^K + \boxed{(K+1) \times 2^{K+1}} = K \cdot 2^{K+2} + 2 \quad \textcircled{1}$$

$$\text{LHS} = (K-1)2^{K+1} + 2 + (K+1)2^{K+1}$$

$$= 2^{K+1}(2K) + 2$$

$$= 2^{K+1} \cdot 2 \times K + 2$$

$$= 2^{K+2} K + 2 \quad \textcircled{1}$$

= RHS. $\therefore$ true for $n=k+1$ if true for $n=k$. $\therefore$ true for $n=1, n=2 \rightarrow$ for all n by $\textcircled{1}$

13. a)(i) $\sqrt{3} \sin x - \cos x = R \sin(x+\alpha)$
 $= R \sin x \cos \alpha + R \cos x \sin \alpha$

$$\therefore R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = -1$$

$$\therefore \tan \alpha = \frac{-1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{5\pi}{6}, \frac{11\pi}{6}$$

but $R > 0$, $\cos \alpha > 0$, $\sin \alpha < 0$

$\therefore \alpha$ in Quad 4. $\therefore \alpha = \frac{5\pi}{6}$

$$R^2 = (-1)^2 + (\sqrt{3})^2$$

$$= 4$$

$$R = 2$$

$$\therefore \sqrt{3} \sin x - \cos x = 2 \sin(x + \frac{5\pi}{6})$$

(ii) $\therefore$ Least value of $\sqrt{3} \sin x - \cos x = -2$

$$\sin(x + \frac{11\pi}{6}) = -1$$

$$x + \frac{11\pi}{6} = \frac{3\pi}{2}, \frac{7\pi}{2}$$

but $x > 0$ $x = \frac{21\pi}{6} - \frac{11\pi}{6} \Rightarrow x = \frac{5\pi}{3}$

13b)

$$\alpha + \beta + \gamma = \frac{2K-4}{3}$$

$$\alpha \beta \gamma = -\frac{K^2}{3}$$

$$\therefore 2K-4 = -2K^2$$

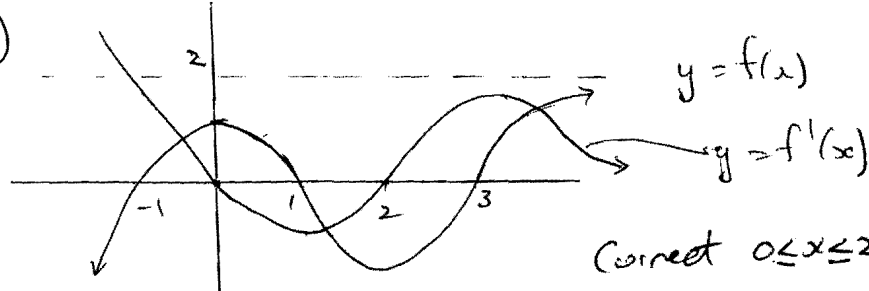
13c) $E \dots I$ $2K^2 + 2K - 4 = 0$

(1) 3 letters between E & I in P brings $K^2 + K - 2 = 0$

ways: $60 \times 3 \times 2$ (4) correct $(K+2)(K-1) = 0$

ways: (1) paper $\therefore K = -2, 1$

13d)



Correct $0 \leq x \leq 2$

Correct the rest

$x < 0$ $x > 2$

e) $\frac{dP}{dt} = KP(L-P)$

i) $P = \frac{LC}{C + e^{-KLt}} = LC(C + e^{-KLt})^{-1}$

$$\frac{dP}{dt} = -LC(C + e^{-KLt})^{-2} \cdot -KL e^{-KLt}$$

$$= \frac{KL^2 C e^{-KLt}}{(C + e^{-KLt})^2}$$

CONSIDER: $\frac{dP}{dt} = KP(L-P)$

$$= K \cdot \frac{LC}{C + e^{-KLt}} \left(L - \frac{LC}{C + e^{-KLt}} \right)$$

$$= \frac{KL^2 C}{(C + e^{-KLt})^2} \frac{LC + LC e^{-KLt} - LC}{C + e^{-KLt}}$$

① - Differentiating further progress

① fully correct.

$$= \frac{KL^2 C e^{-KLt}}{(C + e^{-KLt})^2} = \frac{dP}{dt}$$

ii) as $t \rightarrow \infty$ $P \rightarrow L$

14a) by SIM Δ 's

i) $\frac{r}{30} = \frac{h}{30}$ (matching sides in III Δ 's) $\Rightarrow r = h$

$V = \frac{1}{3} \pi r^2 h$
 $= \frac{1}{3} \pi h^2 \cdot h$
 $= \frac{1}{3} \pi h^3$

ii) $\frac{dV}{dt} = \frac{dh}{dt} \cdot \frac{dV}{dh}$
 $24 = \frac{dh}{dt} \cdot \pi h^2$ (1)
 $h = 16$
 $\frac{dh}{dt} = \frac{24^3}{\pi \cdot 16^3}$
 $= \frac{3}{32\pi} \text{ cm/s}$ (1)

iii) $S = \pi r^2$
 $= \pi h^2$
 $\frac{dS}{dt} = \frac{dS}{dh} \cdot \frac{dh}{dt}$
 $= 2\pi h \cdot \frac{3}{32\pi}$ (1)
 $= 3 \text{ cm}^2/\text{s}$

$\ddot{x} = 8x(x^2+1)$

iv) $\frac{1}{2} v^2 = \int 8x^3 + 8x dx$
 $\frac{1}{2} v^2 = \frac{8x^4}{4} + 4x^2 + C$ (1)
 $x = 0, v = -2$
 $2 = C$
 $\frac{1}{2} v^2 = 2x^4 + 4x^2 + 2$
 $v^2 = 4x^4 + 8x^2 + 4$
 $= 4(x^4 + 2x^2 + 1)$
 $v^2 = 4(x^2 + 1)^2$
 $v = \pm 2(x^2 + 1)$
 $\text{speed} = 2(x^2 + 1) \text{ cm/s}$ (1)

iii) $v = \pm 2(x^2 + 1)$
 but when $x = 0, v = -2$.
 $\therefore v = -2(x^2 + 1)$ (1)
 $\frac{dx}{dt} = -2(x^2 + 1)$
 $\frac{dt}{dx} = \frac{1}{-2(x^2 + 1)}$
 $t = -\frac{1}{2} \int \frac{1}{x^2 + 1} dx$
 $t = -\frac{1}{2} \tan^{-1}(x) + C$
 $t = 0, x = 0, \therefore C = 0$
 $t = -\frac{1}{2} \tan^{-1}(x)$
 $-2t = \tan^{-1}(x) \Rightarrow x = \tan(-2t)$ (1)
 $= -\tan(2t)$

14c) $x = vt \cos \theta$

$\therefore t = \frac{x}{v \cos \theta}$

$y = -\frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2 + \frac{vx \sin \theta}{v \cos \theta}$ simplify
 $= -\frac{gx^2}{2v^2 \cos^2 \theta} + x \tan \theta$ (1)

$= x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$
 $= x \tan \theta - \frac{gx^2}{2v^2} (1 + \tan^2 \theta)$

clearly slow relationship (1)

$\frac{gx^2}{v^2} \propto \frac{1}{h}$ but $v^2 = 2gh$

$y = x \tan \theta - \frac{gx^2}{4gh} (1 + \tan^2 \theta)$
 $= x \tan \theta - \frac{x^2}{4h} (1 + \tan^2 \theta)$ (1)

ii) If Projectile to Pass thru (X, Y)

$$Y = X \tan \theta - \frac{X^2}{4h} (1 + \tan^2 \theta)$$

$$\therefore X^2 \tan^2 \theta - 4hX \tan \theta + (4hY + X^2) = 0. \quad (1)$$

FOR DIFFERENT ROOTS $\Delta > 0$

$$\therefore 16h^2 X^2 - 4X^2(4hY + X^2) > 0 \quad \&$$

$$4X^2(4h^2 - 4hY - X^2) > 0$$

$$\therefore \text{Since } 4X^2 > 0, \quad 4h^2 - 4hY - X^2 > 0$$

$$\therefore 4h(h - Y) > X^2. \quad (1)$$

identifies $\tan \theta$ as the variable & $Y \neq 0$

uses $\Delta > 0$ to find 2 solutions

iii) If $\tan \theta_1, \tan \theta_2$ are roots of quadratic eqn

$$X^2 \tan^2 \theta - 4hX \tan \theta + (4hY + X^2) = 0$$

$$\therefore \tan \theta_1 \cdot \tan \theta_2 = \frac{4hY + X^2}{X^2}$$
$$= 1 + \frac{4hY}{X^2}$$

$$\therefore \tan \theta_1 \text{ or } \tan \theta_2 > 1$$
$$\therefore \theta_1 \text{ or } \theta_2 > \pi/4.$$

(1)